

3.8 Exercises

Exercise 3.1 In terms of the $\hat{x}_s, \hat{y}_s, \hat{z}_s$ coordinates of a fixed space frame $\{s\}$, the frame $\{a\}$ has its $\hat{x}_a$ -axis pointing in the direction $(0, 0, 1)$ and its $\hat{y}_a$ -axis pointing in the direction $(-1, 0, 0)$, and the frame $\{b\}$ has its $\hat{x}_b$ -axis pointing in the direction $(1, 0, 0)$ and its $\hat{y}_b$ -axis pointing in the direction $(0, 0, -1)$.

- Draw by hand the three frames, at different locations so that they are easy to see.
- Write down the rotation matrices R_{sa} and R_{sb} .
- Given R_{sb} , how do you calculate R_{sb}^{-1} without using a matrix inverse? Write down R_{sb}^{-1} and verify its correctness using your drawing.
- Given R_{sa} and R_{sb} , how do you calculate R_{ab} (again without using matrix inverses)? Compute the answer and verify its correctness using your drawing.
- Let $R = R_{sb}$ be considered as a transformation operator consisting of a rotation about $\hat{x}$ by -90° . Calculate $R_1 = R_{sa}R$, and think of R_{sa} as a representation of an orientation, R as a rotation of R_{sa} , and R_1 as the new orientation after the rotation has been performed. Does the new orientation R_1 correspond to a rotation of R_{sa} by -90° about the world-fixed $\hat{x}_s$ -axis or about the body-fixed $\hat{x}_a$ -axis? Now calculate $R_2 = RR_{sa}$. Does the new orientation R_2 correspond to a rotation of R_{sa} by -90° about the world-fixed $\hat{x}_s$ -axis or about the body-fixed $\hat{x}_a$ -axis?
- Use R_{sb} to change the representation of the point $p_b = (1, 2, 3)$ (which is in $\{b\}$ coordinates) to $\{s\}$ coordinates.
- Choose a point p represented by $p_s = (1, 2, 3)$ in $\{s\}$ coordinates. Calculate $p' = R_{sb}p_s$ and $p'' = R_{sb}^T p_s$. For each operation, should the result be interpreted as changing coordinates (from the $\{s\}$ frame to $\{b\}$) without moving the point p or as moving the location of the point without changing the reference frame of the representation?
- An angular velocity w is represented in $\{s\}$ as $\omega_s = (3, 2, 1)$. What is its representation ω_a in $\{a\}$?
- By hand, calculate the matrix logarithm $[\hat{\omega}]\theta$ of R_{sa} . (You may verify your answer with software.) Extract the unit angular velocity $\hat{\omega}$ and rotation amount θ . Redraw the fixed frame $\{s\}$ and in it draw $\hat{\omega}$.
- Calculate the matrix exponential corresponding to the exponential coordinates of rotation $\hat{\omega}\theta = (1, 2, 0)$. Draw the corresponding frame relative to $\{s\}$, as well as the rotation axis $\hat{\omega}$.

Exercise 3.2 Let p be a point whose coordinates are $p = \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{2}}\right)$ with respect to the fixed frame $\hat{x}\text{--}\hat{y}\text{--}\hat{z}$. Suppose that p is rotated about the fixed-frame $\hat{x}$ -axis by 30 degrees, then about the fixed-frame $\hat{y}$ -axis by 135 degrees, and finally about the fixed-frame $\hat{z}$ -axis by -120 degrees. Denote the coordinates of this newly rotated point by p' .

- (a) What are the coordinates p' ?
- (b) Find the rotation matrix R such that $p' = Rp$ for the p' you obtained in (a).

Exercise 3.3 Suppose that $p_i \in \mathbb{R}^3$ and $p'_i \in \mathbb{R}^3$ are related by $p'_i = Rp_i$, $i = 1, 2, 3$, for some unknown rotation matrix R . Find, if it exists, the rotation R for the three input–output pairs $p_i \mapsto p'_i$, where

$$\begin{aligned} p_1 = (\sqrt{2}, 0, 2) &\mapsto p'_1 = (0, 2, \sqrt{2}), \\ p_2 = (1, 1, -1) &\mapsto p'_2 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, -\sqrt{2}\right), \\ p_3 = (0, 2\sqrt{2}, 0) &\mapsto p'_3 = (-\sqrt{2}, \sqrt{2}, -2). \end{aligned}$$

Exercise 3.4 In this exercise you are asked to prove the property $R_{ab}R_{bc} = R_{ac}$ of Equation (3.22). Define the unit axes of frames $\{a\}$, $\{b\}$, and $\{c\}$ by the triplets of orthogonal unit vectors $\{\hat{x}_a, \hat{y}_a, \hat{z}_a\}$, $\{\hat{x}_b, \hat{y}_b, \hat{z}_b\}$, and $\{\hat{x}_c, \hat{y}_c, \hat{z}_c\}$, respectively. Suppose that the unit axes of frame $\{b\}$ can be expressed in terms of the unit axes of frame $\{a\}$ by

$$\begin{aligned} \hat{x}_b &= r_{11}\hat{x}_a + r_{21}\hat{y}_a + r_{31}\hat{z}_a, \\ \hat{y}_b &= r_{12}\hat{x}_a + r_{22}\hat{y}_a + r_{32}\hat{z}_a, \\ \hat{z}_b &= r_{13}\hat{x}_a + r_{23}\hat{y}_a + r_{33}\hat{z}_a. \end{aligned}$$

Similarly, suppose that the unit axes of frame $\{c\}$ can be expressed in terms of the unit axes of frame $\{b\}$ by

$$\begin{aligned} \hat{x}_c &= s_{11}\hat{x}_b + s_{21}\hat{y}_b + s_{31}\hat{z}_b, \\ \hat{y}_c &= s_{12}\hat{x}_b + s_{22}\hat{y}_b + s_{32}\hat{z}_b, \\ \hat{z}_c &= s_{13}\hat{x}_b + s_{23}\hat{y}_b + s_{33}\hat{z}_b. \end{aligned}$$

From the above prove that $R_{ab}R_{bc} = R_{ac}$.

Exercise 3.5 Find the exponential coordinates $\hat{\omega}\theta \in \mathbb{R}^3$ for the $SO(3)$ matrix

$$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}.$$

Exercise 3.6 Given $R = \text{Rot}(\hat{x}, \pi/2)\text{Rot}(\hat{z}, \pi)$, find the unit vector $\hat{\omega}$ and angle θ such that $R = e^{[\hat{\omega}]\theta}$.

Exercise 3.7

(a) Given the rotation matrix

$$R = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix},$$

find all possible values for $\hat{\omega} \in \mathbb{R}^3$, $\|\hat{\omega}\| = 1$, and $\theta \in [0, 2\pi)$ such that $e^{[\hat{\omega}]\theta} = R$.

(b) The two vectors $v_1, v_2 \in \mathbb{R}^3$ are related by

$$v_2 = Rv_1 = e^{[\hat{\omega}]\theta} v_1$$

where $\hat{\omega} \in \mathbb{R}^3$ has length 1, and $\theta \in [-\pi, \pi]$. Given $\hat{\omega} = (\frac{2}{3}, \frac{2}{3}, \frac{1}{3})$, $v_1 = (1, 0, 1)$, $v_2 = (0, 1, 1)$, find all the angles θ that satisfy the above equation.

Exercise 3.8

(a) Suppose that we are seeking the logarithm of a rotation matrix R whose trace is -1 . From the exponential formula

$$e^{[\hat{\omega}]\theta} = I + \sin \theta [\hat{\omega}] + (1 - \cos \theta) [\hat{\omega}]^2, \quad \|\omega\| = 1,$$

and recalling that $\text{tr } R = -1$ implies $\theta = \pi$, the above equation simplifies to

$$R = I + 2[\hat{\omega}]^2 = \begin{bmatrix} 1 - 2(\hat{\omega}_2^2 + \hat{\omega}_3^2) & 2\hat{\omega}_1\hat{\omega}_2 & 2\hat{\omega}_1\hat{\omega}_3 \\ 2\hat{\omega}_1\hat{\omega}_2 & 1 - 2(\hat{\omega}_1^2 + \hat{\omega}_3^2) & 2\hat{\omega}_2\hat{\omega}_3 \\ 2\hat{\omega}_1\hat{\omega}_2 & 2\hat{\omega}_2\hat{\omega}_3 & 1 - 2(\hat{\omega}_1^2 + \hat{\omega}_2^2) \end{bmatrix}.$$

Using the fact that $\hat{\omega}_1^2 + \hat{\omega}_2^2 + \hat{\omega}_3^2 = 1$, is it correct to conclude that

$$\hat{\omega}_1 = \sqrt{\frac{r_{11} + 1}{2}}, \quad \hat{\omega}_2 = \sqrt{\frac{r_{22} + 1}{2}}, \quad \hat{\omega}_3 = \sqrt{\frac{r_{33} + 1}{2}},$$

where r_{ij} denotes the (i, j) th entry of R , is also a solution?

- (b) Using the fact that $[\hat{\omega}]^3 = -[\hat{\omega}]$, the identity $R = I + 2[\hat{\omega}]^2$ can be written in the alternative forms

$$\begin{aligned} R - I &= 2[\hat{\omega}]^2, \\ [\hat{\omega}](R - I) &= 2[\hat{\omega}]^3 = -2[\hat{\omega}], \\ [\hat{\omega}](R + I) &= 0. \end{aligned}$$

The resulting equation consists of three linear equations in $(\hat{\omega}_1, \hat{\omega}_2, \hat{\omega}_3)$. What is the relation between the solution to this linear system and the logarithm of R ?

Exercise 3.9 Exploiting the known properties of rotation matrices, determine the minimum number of arithmetic operations (multiplication and division, addition and subtraction) required to multiply two rotation matrices.

Exercise 3.10 Because arithmetic precision is only finite, the numerically obtained product of two rotation matrices is not necessarily a rotation matrix; that is, the resulting rotation A may not exactly satisfy $A^T A = I$ as desired. Devise an iterative numerical procedure that takes an arbitrary matrix $A \in \mathbb{R}^{3 \times 3}$ and produces a matrix $R \in SO(3)$ that minimizes

$$\|A - R\|^2 = \text{tr}(A - R)(A - R)^T.$$

(Hint: See Appendix D for the relevant background on optimization.)

Exercise 3.11 Properties of the matrix exponential.

- (a) Under what conditions on general $A, B \in \mathbb{R}^{n \times n}$ does $e^A e^B = e^{A+B}$ hold?
 (b) If $A = [\mathcal{V}_a]$ and $B = [\mathcal{V}_b]$, where $\mathcal{V}_a = (\omega_a, v_a)$ and $\mathcal{V}_b = (\omega_b, v_b)$ are arbitrary twists, then under what conditions on $\mathcal{V}_a$ and $\mathcal{V}_b$ does $e^A e^B = e^{A+B}$ hold? Try to give a physical description of this condition.

Exercise 3.12

- (a) Given a rotation matrix $A = \text{Rot}(\hat{z}, \alpha)$, where $\text{Rot}(\hat{z}, \alpha)$ indicates a rotation about the $\hat{z}$ -axis by an angle α , find all rotation matrices $R \in SO(3)$ that satisfy $AR = RA$.
 (b) Given rotation matrices $A = \text{Rot}(\hat{z}, \alpha)$ and $B = \text{Rot}(\hat{z}, \beta)$, with $\alpha \neq \beta$, find all rotation matrices $R \in SO(3)$ that satisfy $AR = RB$.
 (c) Given arbitrary rotation matrices $A, B \in SO(3)$, find all solutions $R \in SO(3)$ to the equation $AR = RB$.

Exercise 3.13

- (a) Show that the three eigenvalues of a rotation matrix $R \in SO(3)$ each have unit magnitude, and conclude that they can always be written $\{\mu + i\nu, \mu - i\nu, 1\}$, where $\mu^2 + \nu^2 = 1$.
- (b) Show that a rotation matrix $R \in SO(3)$ can always be factored in the form

$$R = A \begin{bmatrix} \mu & \nu & 0 \\ -\nu & \mu & 0 \\ 0 & 0 & 1 \end{bmatrix} A^{-1},$$

where $A \in SO(3)$ and $\mu^2 + \nu^2 = 1$. (Hint: Denote the eigenvector associated with the eigenvalue $\mu + i\nu$ by $x + iy$, $x, y \in \mathbb{R}^3$, and the eigenvector associated with the eigenvalue 1 by $z \in \mathbb{R}^3$. For the purposes of this problem you may assume that the set of vectors $\{x, y, z\}$ can always be chosen to be linearly independent.)

Exercise 3.14 Given $\omega \in \mathbb{R}^3$, $\|\omega\| = 1$, and θ a nonzero scalar, show that

$$(I\theta + (1 - \cos \theta)[\omega] + (\theta - \sin \theta)[\omega]^2)^{-1} = \frac{1}{\theta}I - \frac{1}{2}[\omega] + \left(\frac{1}{\theta} - \frac{1}{2} \cot \frac{\theta}{2}\right)[\omega]^2.$$

(Hint: From the identity $[\omega]^3 = -[\omega]$, express the inverse as a quadratic matrix polynomial in $[\omega]$.)

Exercise 3.15

- (a) Given a fixed frame $\{0\}$ and a moving frame $\{1\}$ initially aligned with $\{0\}$, perform the following sequence of rotations on $\{1\}$:
1. Rotate $\{1\}$ about the $\{0\}$ frame $\hat{x}$ -axis by α ; call this new frame $\{2\}$.
 2. Rotate $\{2\}$ about the $\{0\}$ frame $\hat{y}$ -axis by β ; call this new frame $\{3\}$.
 3. Rotate $\{3\}$ about the $\{0\}$ frame $\hat{z}$ -axis by γ ; call this new frame $\{4\}$.
- What is the final orientation R_{04} ?
- (b) Suppose that the third step above is replaced by the following: “Rotate $\{3\}$ about the $\hat{z}$ -axis of frame $\{3\}$ by γ ; call this new frame $\{4\}$.” What is the final orientation R_{04} ?

- (c) Find T_{ca} for the following transformations:

$$T_{ab} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & -1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad T_{cb} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 1 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Exercise 3.16 In terms of the $\hat{x}_s, \hat{y}_s, \hat{z}_s$ coordinates of a fixed space frame $\{s\}$, frame $\{a\}$ has its $\hat{x}_a$ -axis pointing in the direction $(0, 0, 1)$ and its $\hat{y}_a$ -axis pointing in the direction $(-1, 0, 0)$, and frame $\{b\}$ has its $\hat{x}_b$ -axis pointing in the direction $(1, 0, 0)$ and its $\hat{y}_b$ -axis pointing in the direction $(0, 0, -1)$. The origin of $\{a\}$ is at $(3, 0, 0)$ in $\{s\}$ and the origin of $\{b\}$ is at $(0, 2, 0)$ in $\{s\}$.

- Draw by hand a diagram showing $\{a\}$ and $\{b\}$ relative to $\{s\}$.
- Write down the rotation matrices R_{sa} and R_{sb} and the transformation matrices T_{sa} and T_{sb} .
- Given T_{sb} , how do you calculate T_{sb}^{-1} without using a matrix inverse? Write T_{sb}^{-1} and verify its correctness using your drawing.
- Given T_{sa} and T_{sb} , how do you calculate T_{ab} (again without using matrix inverses)? Compute the answer and verify its correctness using your drawing.
- Let $T = T_{sb}$ be considered as a transformation operator consisting of a rotation about $\hat{x}$ by -90° and a translation along $\hat{y}$ by 2 units. Calculate $T_1 = T_{sa}T$. Does T_1 correspond to a rotation and translation about $\hat{x}_s$ and $\hat{y}_s$, respectively (a world-fixed transformation of T_{sa}), or a rotation and translation about $\hat{x}_a$ and $\hat{y}_a$, respectively (a body-fixed transformation of T_{sa})? Now calculate $T_2 = TT_{sa}$. Does T_2 correspond to a body-fixed or world-fixed transformation of T_{sa} ?
- Use T_{sb} to change the representation of the point $p_b = (1, 2, 3)$ in $\{b\}$ coordinates to $\{s\}$ coordinates.
- Choose a point p represented by $p_s = (1, 2, 3)$ in $\{s\}$ coordinates. Calculate $p' = T_{sb}p_s$ and $p'' = T_{sb}^{-1}p_s$. For each operation, should the result be interpreted as changing coordinates (from the $\{s\}$ frame to $\{b\}$) without moving the point p , or as moving the location of the point without changing the reference frame of the representation?
- A twist $\mathcal{V}$ is represented in $\{s\}$ as $\mathcal{V}_s = (3, 2, 1, -1, -2, -3)$. What is its representation $\mathcal{V}_a$ in frame $\{a\}$?
- By hand, calculate the matrix logarithm $[\mathcal{S}]\theta$ of T_{sa} . (You may verify your answer with software.) Extract the normalized screw axis $\mathcal{S}$ and rotation amount θ . Find a $\{q, \hat{s}, h\}$ representation of the screw axis. Redraw the fixed frame $\{s\}$ and in it draw $\mathcal{S}$.

- (j) Calculate the matrix exponential corresponding to the exponential coordinates of rigid-body motion $\mathcal{S}\theta = (0, 1, 2, 3, 0, 0)$. Draw the corresponding frame relative to $\{s\}$, as well as the screw axis $\mathcal{S}$.

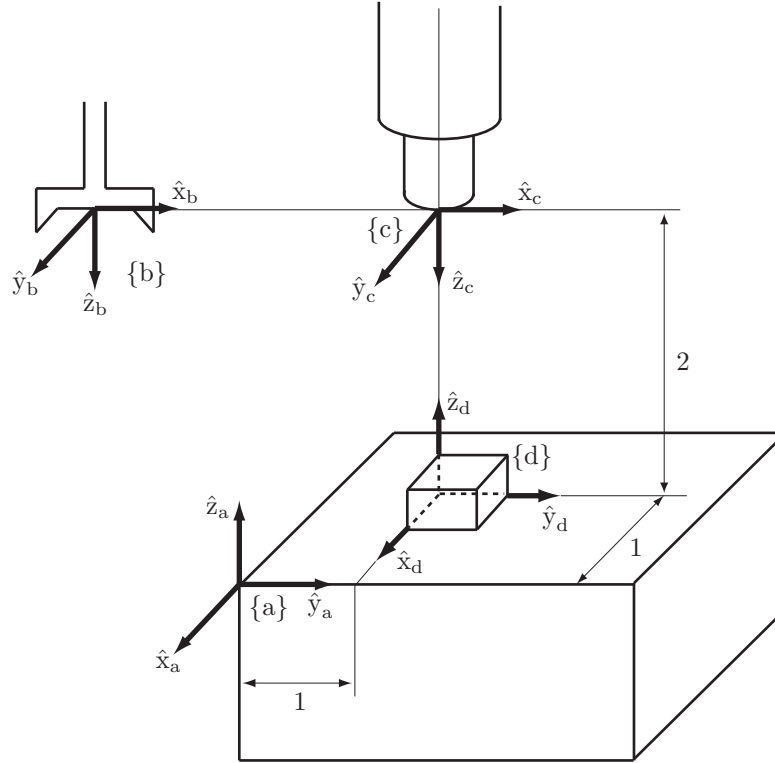


Figure 3.23: Four reference frames defined in a robot's workspace.

Exercise 3.17 Four reference frames are shown in the robot workspace of Figure 3.23: the fixed frame $\{a\}$, the end-effector frame $\{b\}$, the camera frame $\{c\}$, and the workpiece frame $\{d\}$.

- (a) Find T_{ad} and T_{cd} in terms of the dimensions given in the figure.
 (b) Find T_{ab} given that

$$T_{bc} = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

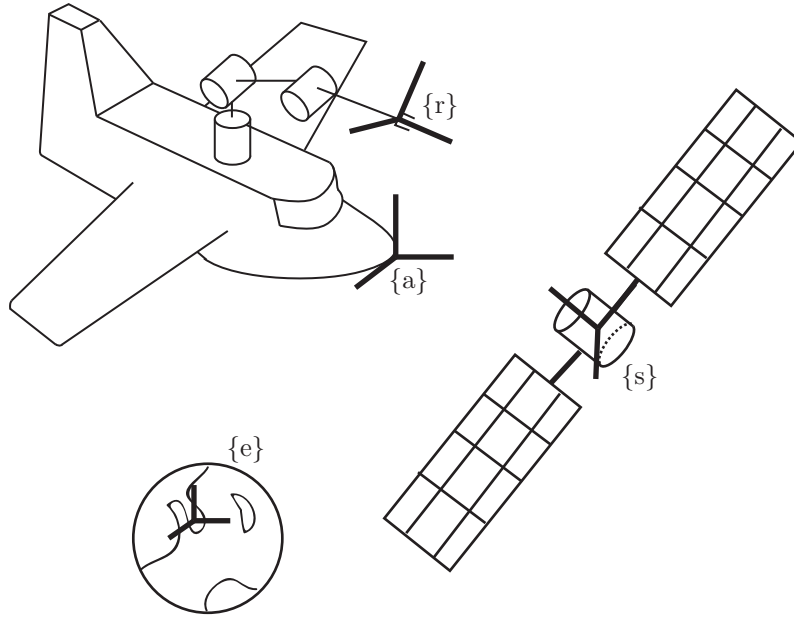


Figure 3.24: A robot arm mounted on a spacecraft.

Exercise 3.18 Consider a robot arm mounted on a spacecraft as shown in Figure 3.24, in which frames are attached to the Earth $\{e\}$, a satellite $\{s\}$, the spacecraft $\{a\}$, and the robot arm $\{r\}$, respectively.

- Given T_{ea} , T_{ar} , and T_{es} , find T_{rs} .
- Suppose that the frame $\{s\}$ origin as seen from $\{e\}$ is $(1, 1, 1)$ and that

$$T_{er} = \begin{bmatrix} -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Write down the coordinates of the frame $\{s\}$ origin as seen from frame $\{r\}$.

Exercise 3.19 Two satellites are circling the Earth as shown in Figure 3.25. Frames $\{1\}$ and $\{2\}$ are rigidly attached to the satellites in such a way that their $\hat{x}$ -axes always point toward the Earth. Satellite 1 moves at a constant speed

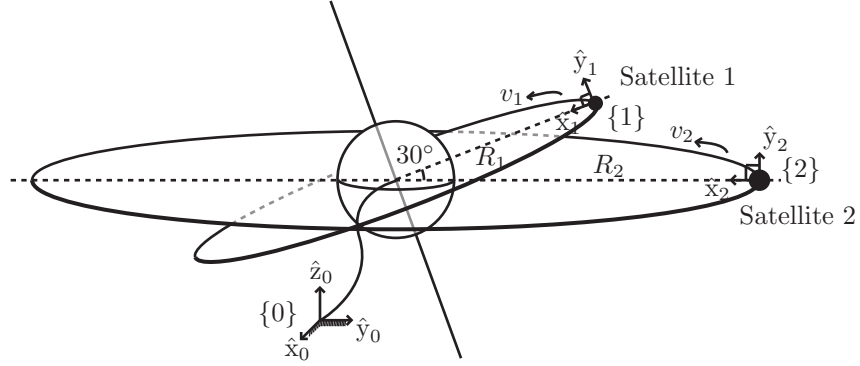


Figure 3.25: Two satellites circling the Earth.

v_1 , while satellite 2 moves at a constant speed v_2 . To simplify matters, ignore the rotation of the Earth about its own axis. The fixed frame $\{0\}$ is located at the center of the Earth. Figure 3.25 shows the position of the two satellites at $t = 0$.

- Derive the frames T_{01} , T_{02} as a function of t .
- Using your results from part (a), find T_{21} as a function of t .

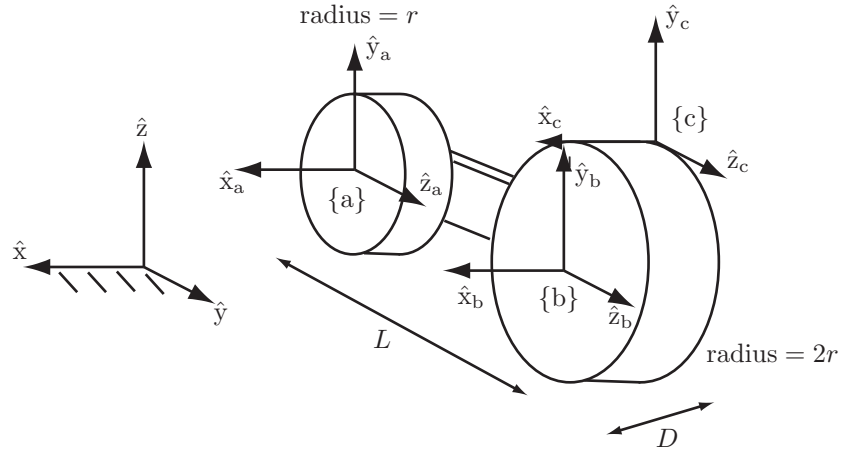


Figure 3.26: A high-wheel bicycle.

Exercise 3.20 Consider the high-wheel bicycle of Figure 3.26, in which the diameter of the front wheel is twice that of the rear wheel. Frames $\{a\}$ and $\{b\}$ are attached respectively to the centers of the wheels, and frame $\{c\}$ is attached to the top of the front wheel. Assuming that the bike moves forward in the $\hat{y}$ -direction, find T_{ac} as a function of the front wheel's rotation angle θ (assume $\theta = 0$ at the instant shown in the figure).

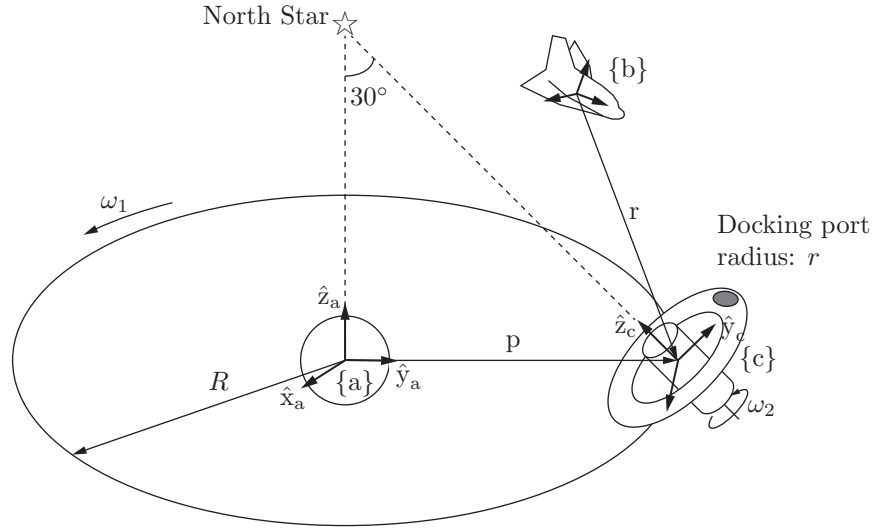


Figure 3.27: A spacecraft and space station.

Exercise 3.21 The space station of Figure 3.27 moves in circular orbit around the Earth, and at the same time rotates about an axis always pointing toward the North Star. Owing to an instrument malfunction, a spacecraft heading toward the space station is unable to locate the docking port. An Earth-based ground station sends the following information to the spacecraft:

$$T_{ab} = \begin{bmatrix} 0 & -1 & 0 & -100 \\ 1 & 0 & 0 & 300 \\ 0 & 0 & 1 & 500 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad p_a = \begin{bmatrix} 0 \\ 800 \\ 0 \end{bmatrix},$$

where p_a is the vector p expressed in $\{a\}$ -frame coordinates.

- From the given information, find r_b , the vector r expressed in $\{b\}$ -frame coordinates.
- Determine T_{bc} at the instant shown in the figure. Assume here that the $\hat{y}$ - and $\hat{z}$ -axes of the $\{a\}$ and $\{c\}$ frames are coplanar with the docking port.

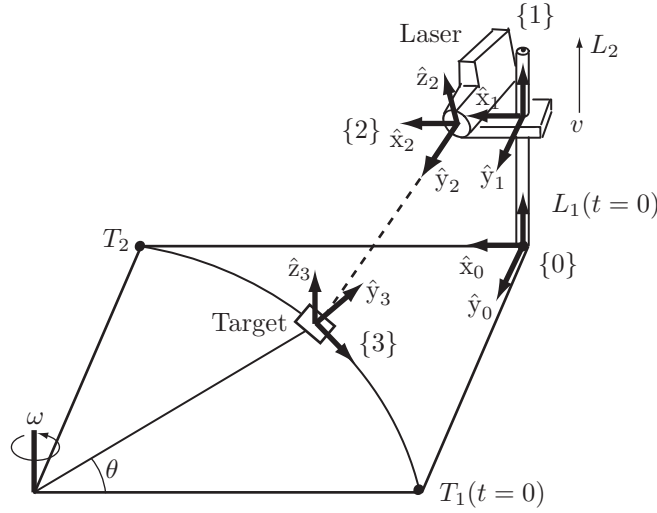


Figure 3.28: A laser tracking a moving target.

Exercise 3.22 A target moves along a circular path at constant angular velocity ω rad/s in the $\hat{x}$ - $\hat{y}$ -plane, as shown in Figure 3.28. The target is tracked by a laser mounted on a moving platform, rising vertically at constant speed v . Assume that at $t = 0$ the laser and the platform start at L_1 , while the target starts at frame T_1 .

- Derive the frames T_{01}, T_{12}, T_{03} as functions of t .
- Using your results from part (a), derive T_{23} as a function of t .

Exercise 3.23 Two toy cars are moving on a round table as shown in Figure 3.29. Car 1 moves at a constant speed v_1 along the circumference of the table, while car 2 moves at a constant speed v_2 along a radius; the positions of the two vehicles at $t = 0$ are shown in the figures.

- Find T_{01} and T_{02} as a function of t .
- Find T_{12} as a function of t .

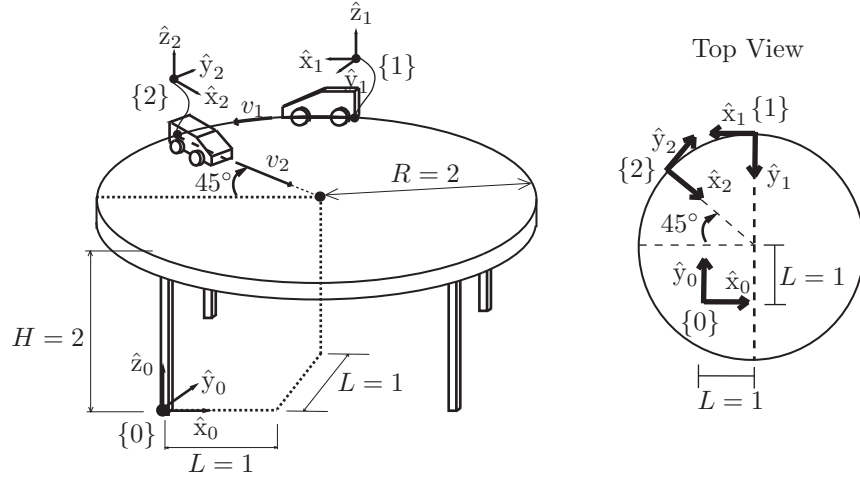


Figure 3.29: Two toy cars on a round table.

Exercise 3.24 Figure 3.30 shows the configuration, at $t = 0$, of a robot arm whose first joint is a screw joint of pitch $h = 2$. The arm's link lengths are $L_1 = 10$, $L_2 = L_3 = 5$, and $L_4 = 3$. Suppose that all joint angular velocities are constant, with values $\omega_1 = \pi/4$, $\omega_2 = \pi/8$, $\omega_3 = -\pi/4$ rad/s. Find $T_{sb}(4) \in SE(3)$, i.e., the configuration of the end-effector frame $\{b\}$ relative to the fixed frame $\{s\}$ at time $t = 4$.

Exercise 3.25 A camera is rigidly attached to a robot arm, as shown in Figure 3.31. The transformation $X \in SE(3)$ is constant. The robot arm moves from posture 1 to posture 2. The transformations $A \in SE(3)$ and $B \in SE(3)$ are measured and can be assumed to be known.

(a) Suppose that X and A are given as follows:

$$X = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

What is B ?

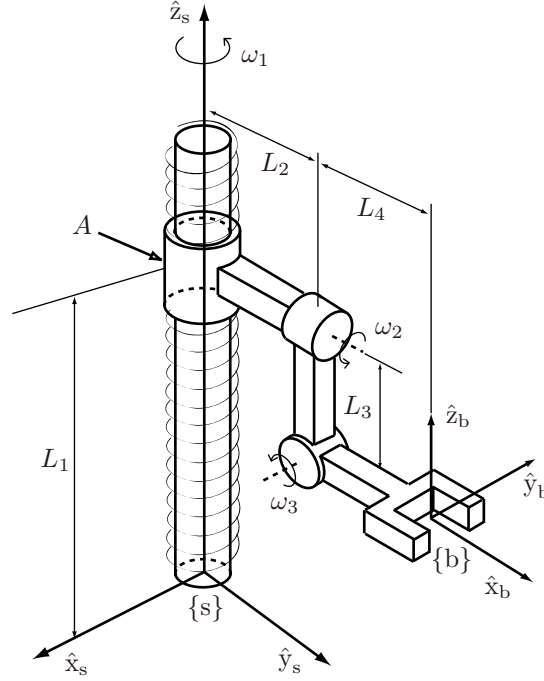


Figure 3.30: A robot arm with a screw joint.

(b) Now suppose that

$$A = \begin{bmatrix} R_A & p_A \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} R_B & p_B \\ 0 & 1 \end{bmatrix}$$

are known and we wish to find

$$X = \begin{bmatrix} R_X & p_X \\ 0 & 1 \end{bmatrix}.$$

Set $R_A = e^{[\alpha]}$ and $R_B = e^{[\beta]}$. What are the conditions on $\alpha \in \mathbb{R}^3$ and $\beta \in \mathbb{R}^3$ for a solution R_X to exist?

(c) Now suppose that we have a set of k equations

$$A_i X = X B_i \quad \text{for } i = 1, \dots, k.$$

Assume that A_i and B_i are all known. What is the minimum number k for which a unique solution exists?

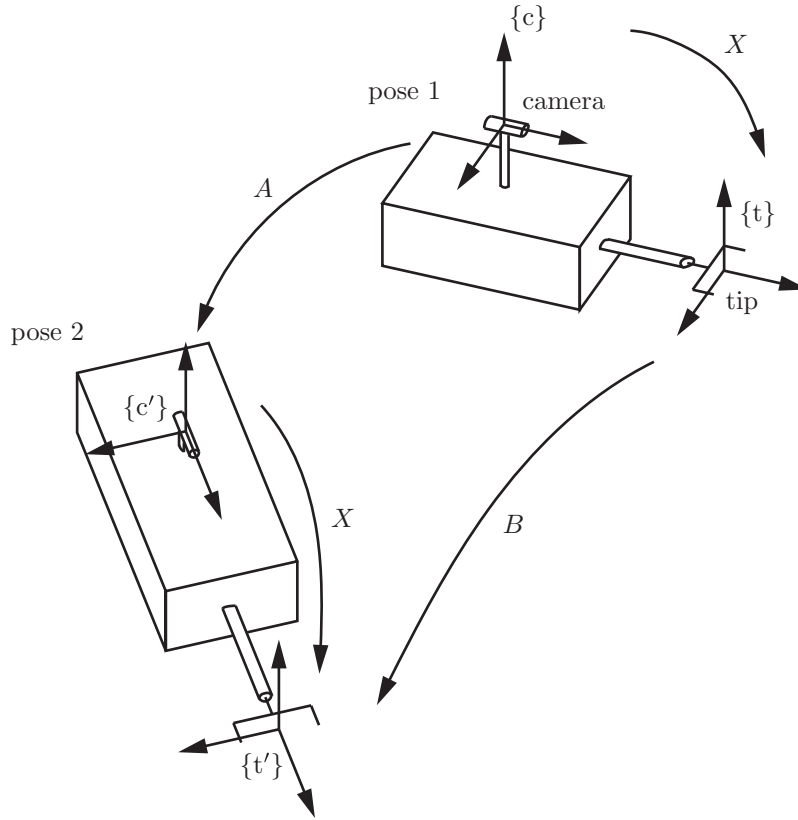


Figure 3.31: A camera rigidly attached to a robot arm.

Exercise 3.26 Draw the screw axis for which $q = (3, 0, 0)$, $\hat{s} = (0, 0, 1)$, and $h = 2$.

Exercise 3.27 Draw the screw axis for the twist $\mathcal{V} = (0, 2, 2, 4, 0, 0)$.

Exercise 3.28 Assume that the space-frame angular velocity is $\omega_s = (1, 2, 3)$

for a moving body with frame $\{b\}$ at

$$R = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

relative to the space frame $\{s\}$. Calculate the body's angular velocity ω_b in $\{b\}$.

Exercise 3.29 Two frames $\{a\}$ and $\{b\}$ are attached to a moving rigid body. Show that the twist of $\{a\}$ in space-frame coordinates is the same as the twist of $\{b\}$ in space-frame coordinates.

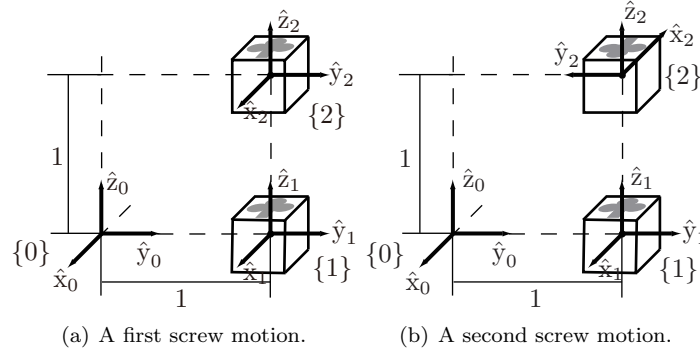


Figure 3.32: A cube undergoing two different screw motions.

Exercise 3.30 A cube undergoes two different screw motions from frame $\{1\}$ to frame $\{2\}$ as shown in Figure 3.32. In both cases, (a) and (b), the initial configuration of the cube is

$$T_{01} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

- For each case, (a) and (b), find the exponential coordinates $\mathcal{S}\theta = (\omega, v)\theta$ such that $T_{02} = e^{[S]\theta}T_{01}$, where no constraints are placed on ω or v .
- Repeat (a), this time with the constraint that $\|\omega\theta\| \in [-\pi, \pi]$.

Exercise 3.31 In Example 3.19 and Figure 3.16, the block that the robot must pick up weighs 1 kg, which means that the robot must provide approximately

10 N of force in the $\hat{z}_e$ -direction of the block's frame $\{e\}$ (which you can assume is at the block's center of mass). Express this force as a wrench $\mathcal{F}_e$ in the $\{e\}$ frame. Given the transformation matrices in Example 3.19, express this same wrench in the end-effector frame $\{c\}$ as $\mathcal{F}_c$.

Exercise 3.32 Given two reference frames $\{a\}$ and $\{b\}$ in physical space, and a fixed frame $\{o\}$, define the distance between frames $\{a\}$ and $\{b\}$ as

$$\text{dist}(T_{oa}, T_{ob}) \equiv \sqrt{\theta^2 + \|p_{ab}\|^2}$$

where $R_{ab} = e^{[\hat{\omega}]\theta}$. Suppose that the fixed frame is displaced to another frame $\{o'\}$ and that $T_{o'a} = ST_{oa}$, $T_{o'b} = ST_{ob}$ for some constant $S = (R_s, p_s) \in SE(3)$.

(a) Evaluate $\text{dist}(T_{o'a}, T_{o'b})$ using the above distance formula.

(b) Under what conditions on S does $\text{dist}(T_{oa}, T_{ob}) = \text{dist}(T_{o'a}, T_{o'b})$?

Exercise 3.33 (a) Find the general solution to the differential equation $\dot{x} = Ax$, where

$$A = \begin{bmatrix} -2 & 1 \\ 0 & -1 \end{bmatrix}.$$

What happens to the solution $x(t)$ as $t \rightarrow \infty$?

(b) Do the same for

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}.$$

What happens to the solution $x(t)$ as $t \rightarrow \infty$?

Exercise 3.34 Let $x \in \mathbb{R}^2$, $A \in \mathbb{R}^{2 \times 2}$, and consider the linear differential equation $\dot{x}(t) = Ax(t)$. Suppose that

$$x(t) = \begin{bmatrix} e^{-3t} \\ -3e^{-3t} \end{bmatrix}$$

is a solution for the initial condition $x(0) = (1, -3)$, and

$$x(t) = \begin{bmatrix} e^t \\ e^t \end{bmatrix}$$

is a solution for the initial condition $x(0) = (1, 1)$. Find A and e^{At} .

Exercise 3.35 Given a differential equation of the form $\dot{x} = Ax + f(t)$, where $x \in \mathbb{R}^n$ and $f(t)$ is a given differentiable function of t , show that the general

solution can be written

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-s)}f(s) ds.$$

(Hint: Define $z(t) = e^{-At}x(t)$ and evaluate $\dot{z}(t)$.)

Exercise 3.36 Referring to Appendix B, answer the following questions related to ZYZ Euler angles.

- Derive a procedure for finding the ZYZ Euler angles of a rotation matrix.
- Using the results of (a), find the ZYZ Euler angles for the following rotation matrix:

$$\begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Exercise 3.37 Consider a wrist mechanism with two revolute joints θ_1 and θ_2 , in which the end-effector frame orientation $R \in SO(3)$ is given by

$$R = e^{[\hat{\omega}_1]\theta_1} e^{[\hat{\omega}_2]\theta_2},$$

with $\hat{\omega}_1 = (0, 0, 1)$ and $\hat{\omega}_2 = (0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$. Determine whether the following orientation is reachable (that is, find, if it exists, a solution (θ_1, θ_2) for the following R):

$$R = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Exercise 3.38 Show that rotation matrices of the form

$$\begin{bmatrix} r_{11} & r_{12} & 0 \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

can be parametrized using just two parameters θ and ϕ as follows:

$$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \end{bmatrix}.$$

What should the range of values be for θ and ϕ ?

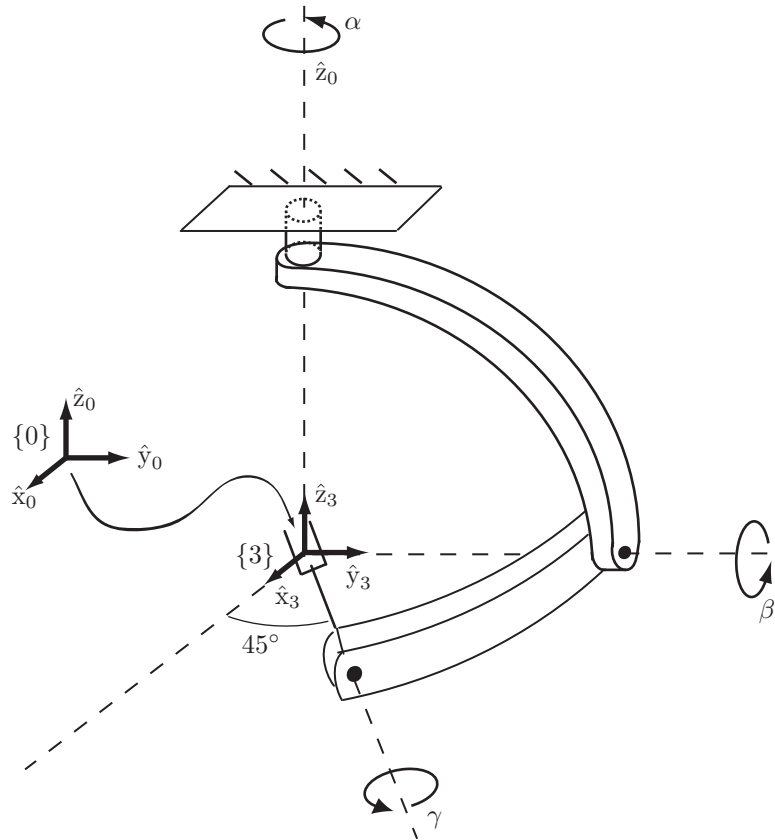


Figure 3.33: A three-degree-of-freedom wrist mechanism.

Exercise 3.39 Figure 3.33 shows a three-dof wrist mechanism in its zero position (i.e., all joint angles are set to zero).

- Express the tool-frame orientation $R_{03} = R(\alpha, \beta, \gamma)$ as a product of three rotation matrices.
- Find all possible angles (α, β, γ) for the two values of R_{03} given below. If no solution exists, explain why this is so in terms of the analogy between $SO(3)$ and a solid ball of radius π .

- (i) $R_{03} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$
- (ii) $R_{03} = e^{[\hat{\omega}]\pi/2},$ where $\hat{\omega} = (0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}).$

Exercise 3.40 Refer to Appendix B.

- Verify formulas (B.10) and (B.11) for the unit quaternion representation of a rotation $R \in SO(3).$
- Verify formula (B.12) for the rotation matrix R representation of a unit quaternion $q \in S^3.$
- Verify the product rule for two unit quaternions. That is, given two unit quaternions $q, p \in S^3$ corresponding respectively to the rotations $R, Q \in SO(3),$ find a formula for the unit quaternion representation of the product $RQ \in SO(3).$

Exercise 3.41 The Cayley transform of Equation (B.18) in Appendix B can be generalized to higher orders as follows:

$$R = (I - [r])^k (I + [r])^{-k}. \quad (3.99)$$

- For the case $k = 2,$ show that the rotation R corresponding to r can be computed from the formula

$$R = I - 4 \frac{1 - r^T r}{(1 + r^T r)^2} [r] + \frac{8}{(1 + r^T r)^2} [r]^2. \quad (3.100)$$

- Conversely, given a rotation matrix $R,$ show that a vector r that satisfies Equation (3.100) can be obtained as

$$r = -\hat{\omega} \tan \frac{\theta}{4}, \quad (3.101)$$

where, as before, $\hat{\omega}$ is the unit vector along the axis of rotation for $R,$ and θ is the corresponding rotation angle. Is this solution unique?

- Show that the angular velocity in the body frame obeys the following relation:

$$\dot{r} = \frac{1}{4} ((1 - r^T r)I + 2[r] + 2rr^T) \omega. \quad (3.102)$$

- Explain what happens to the singularity at π that exists for the standard Cayley–Rodrigues parameters. Discuss the relative advantages and disadvantages of the modified Cayley–Rodrigues parameters, particularly for order $k = 4$ and higher.

- (e) Compare the number of arithmetic operations needed for multiplying two rotation matrices, two unit quaternions, or two Cayley–Rodrigues representations. Which requires the fewest arithmetic operations?

Exercise 3.42 Rewrite the software for Chapter 3 in your favorite programming language.

Exercise 3.43 Write a function that returns “true” if a given 3×3 matrix is within ϵ of being a rotation matrix and “false” otherwise. It is up to you how to define the “distance” between a random 3×3 real matrix and the closest member of $SO(3)$. If the function returns “true,” it should also return the “nearest” matrix in $SO(3)$. See, for example, Exercise 3.10.

Exercise 3.44 Write a function that returns “true” if a given 4×4 matrix is within ϵ of an element of $SE(3)$ and “false” otherwise.

Exercise 3.45 Write a function that returns “true” if a given 3×3 matrix is within ϵ of an element of $so(3)$ and “false” otherwise.

Exercise 3.46 Write a function that returns “true” if a given 4×4 matrix is within ϵ of an element of $se(3)$ and “false” otherwise.

Exercise 3.47 The primary purpose of the provided software is to be easy to read and educational, reinforcing the concepts in the book. The code is optimized neither for efficiency nor robustness, nor does it do full error-checking on its inputs.

Familiarize yourself with the whole code in your favorite language by reading the functions and their comments. This should help cement your understanding of the material in this chapter. Then:

- (a) Rewrite one function to do full error-checking on its input, and have the function return a recognizable error value if the function is called with an improper input (e.g., an argument to the function is not an element of $SO(3)$, $SE(3)$, $so(3)$, or $se(3)$, as expected).
- (b) Rewrite one function to improve its computational efficiency, perhaps by using what you know about properties of rotation or transformation matrices.
- (c) Can you reduce the numerical sensitivity of either of the matrix logarithm functions?

Exercise 3.48 Use the provided software to write a program that allows the user to specify an initial configuration of a rigid body by T , a screw axis specified by $\{q, \hat{s}, h\}$ in the fixed frame $\{s\}$, and the total distance traveled along the screw axis, θ . The program should calculate the final configuration $T_1 = e^{[S]\theta}T$ attained when the rigid body follows the screw S a distance θ , as well as the intermediate configurations at $\theta/4$, $\theta/2$, and $3\theta/4$. At the initial, intermediate, and final configurations, the program should plot the $\{b\}$ axes of the rigid body. The program should also calculate the screw axis S_1 and the distance θ_1 following S_1 that takes the rigid body from T_1 to the origin and it should plot the screw axis S_1 . Test the program with $q = (0, 2, 0)$, $\hat{s} = (0, 0, 1)$, $h = 2$, $\theta = \pi$, and

$$T = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Exercise 3.49 In this chapter, we developed expressions for the matrix exponential for spatial motions mapping elements of $so(3)$ to $SO(3)$ and elements of $se(3)$ to $SE(3)$. Similarly, we developed algorithms for the matrix logarithm going the other direction.

We could also develop matrix exponentials for planar motions, from $so(2)$ to $SO(2)$ and from $se(2)$ to $SE(2)$, as well as the matrix logarithms going from $SO(2)$ to $so(2)$ and $SE(2)$ to $se(2)$. For the $so(2)$ to $SO(2)$ case there is a single exponential coordinate. For the $se(2)$ to $SE(2)$ case there are three exponential coordinates, corresponding to a twist with three elements set to zero, $\mathcal{V} = (0, 0, \omega_z, v_x, v_y, 0)$.

For planar rotations and planar twists we could apply the matrix exponentials and logarithms that we derived for the spatial case by simply expressing the $so(2)$, $SO(2)$, $se(2)$, and $SE(2)$ elements as elements of $so(3)$, $SO(3)$, $se(3)$, and $SE(3)$. Instead, in this problem, write down explicitly the matrix exponential and logarithm for the $so(2)$ to $SO(2)$ case using a single exponential coordinate, and the matrix exponential and logarithm for the $se(2)$ to $SE(2)$ case using three exponential coordinates. Then provide software implementations of each of the four functions in your favorite programming language, and provide execution logs that show that they function as expected.